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BUILDING A CULTURE OF INNOVATION FOR GROWTH

Gypsum False Ceiling

Live beautiful

False Ceiling – Gypsum

Gypsum is a soft, lightweight hydrated sulfate of calcium. It is a widely used material for false ceilings due to the following reasons:

- Gypsum boards are suspended from a **galvanized iron (GI)** framework to construct false ceilings. The tapered edges of the boards allow easier jointing and help achieve a smooth, seamless finish.
- Gypsum is naturally available in shades of **brown, white, red, yellow,** and **grey,** and is also available in a colorless form suitable for surface finishing and painting.
- Gypsum is a flexible material that allows the creation of various shapes and textures, including plain, carved, stepped, curved, and coffered ceiling designs.

A gypsum false ceiling is a dry construction ceiling system made using:

- Gypsum boards (usually 12.5 mm thick)
- GI framework (channels, sections, hangers)
- Jointing compounds + tape
- Fixed to the RCC slab using hangers



False Ceiling

- 2 TYPES OF GYPSUM FALSE CEILING (BY FORM)



1.Plain Gypsum Ceiling



2. Designer Gypsum Ceiling

Note:- All the images are only for reference purposes.

1a. Plain Gypsum Ceiling (Full Horizontal Ceiling)

- **Calculation of gypsum false ceiling (Plain type)**

Plain Flat Ceiling

This type includes:

- **A single-level flat gypsum ceiling**
- **No steps**
- **No cove**

Step 1: Calculate total ceiling area

For a plain flat ceiling, the calculation is **direct**.

Formula:

Length × Width

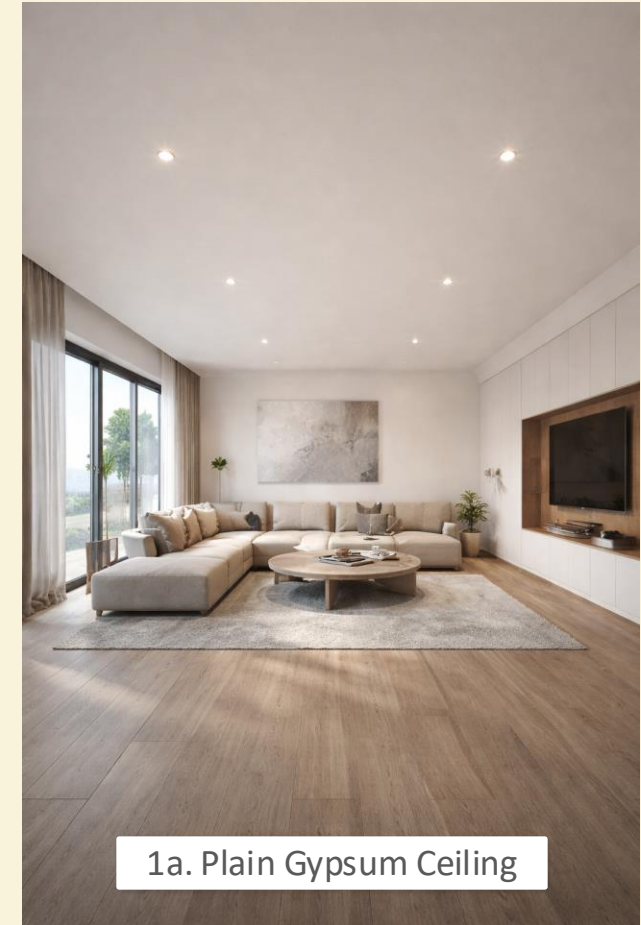
Example:

Room size = **12 ft × 10 ft**

Total ceiling area =

$12 \times 10 = \mathbf{120\ sq.ft}$

This **120 sq.ft** is the **final and only quantity** used for costing.



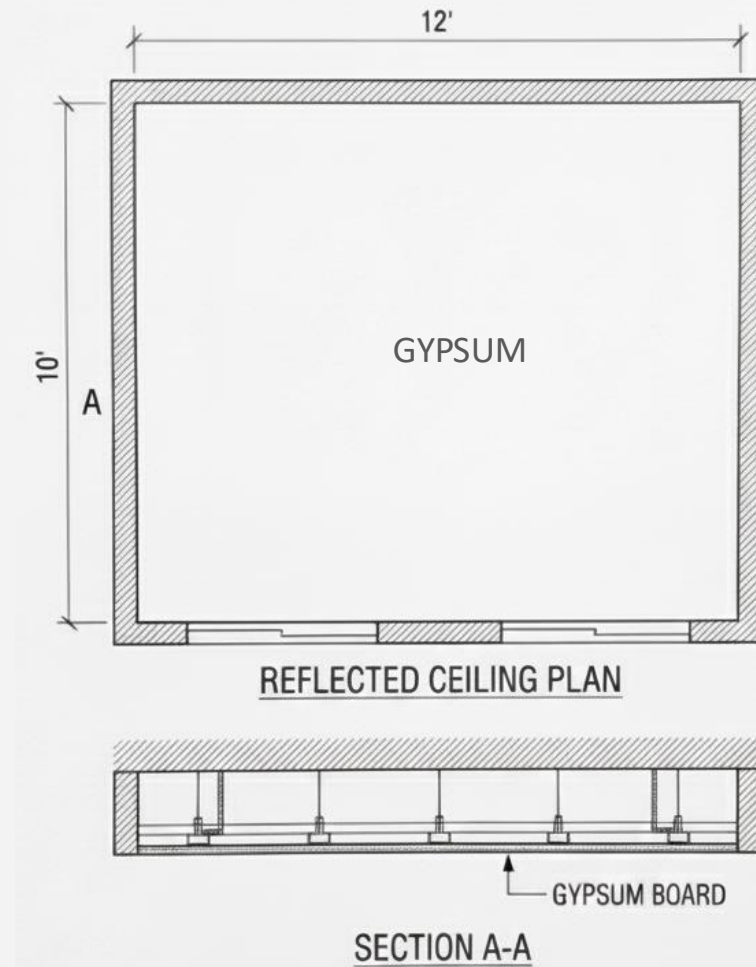
Step 2: Cost calculation (Gypsum only)

(Example rate – adjust as per company standard)

- Gypsum false ceiling (material + GI framework + finishing):
₹110 / sq.ft

Calculation:

$$120 \times 110 = \text{₹13,200}$$



Note: For plain flat gypsum false ceilings, the ceiling area is calculated directly from the room dimensions, and the cost is based solely on the total plan area in square feet.

1b. Plain Gypsum Ceiling (With Cove)

- Calculation of gypsum false ceiling (Rectangular type)

Step 1: First, calculate the **total ceiling area of the room**.

Formula:

Length × Width

Example:

Room size = 12 ft × 10 ft

Total ceiling area =
12 × 10 = **120 sq.ft**

This area is used to calculate:

- Gypsum board, GI framework, Labour, Jointing and finishing

Step 2: In a rectangular ceiling, usually:

- The **outer portion** runs along the walls
- The **inner portion** is dropped or highlighted

This design **does not change the total area**.

Example:

- Inner rectangle offset = 1 ft from all sides
- Inner size = 10 ft × 8 ft
- Inner area = 80 sq.ft
- Outer border area = (120 – 80) = 40 sq.ft**



1b. Plain Gypsum Ceiling

Important Note – Plain Gypsum False Ceiling

In Plain False Ceiling, **maximum 2 vertical steps** are allowed.

Step 3: Cove is provided along the **inner rectangle edges**.

Inner rectangle size = 10 ft × 8 ft

Cove length =

$$10 + 10 + 8 + 8 = \mathbf{36 \text{ running feet}}$$

So:

- Cove lighting length = **36 RFT**

Step 4: Simple **cost example**

(Example rates only)

- Gypsum false ceiling: ₹120 / sq.ft
- Cove groove (gypsum): ₹90 / RFT
- LED strip: ₹120 / RFT (From Electrical SKU)

Calculation:

Gypsum ceiling cost: $40 \times 120 = \mathbf{₹4,800}$

Cove groove cost: $36 \times 90 = \mathbf{₹3,240}$

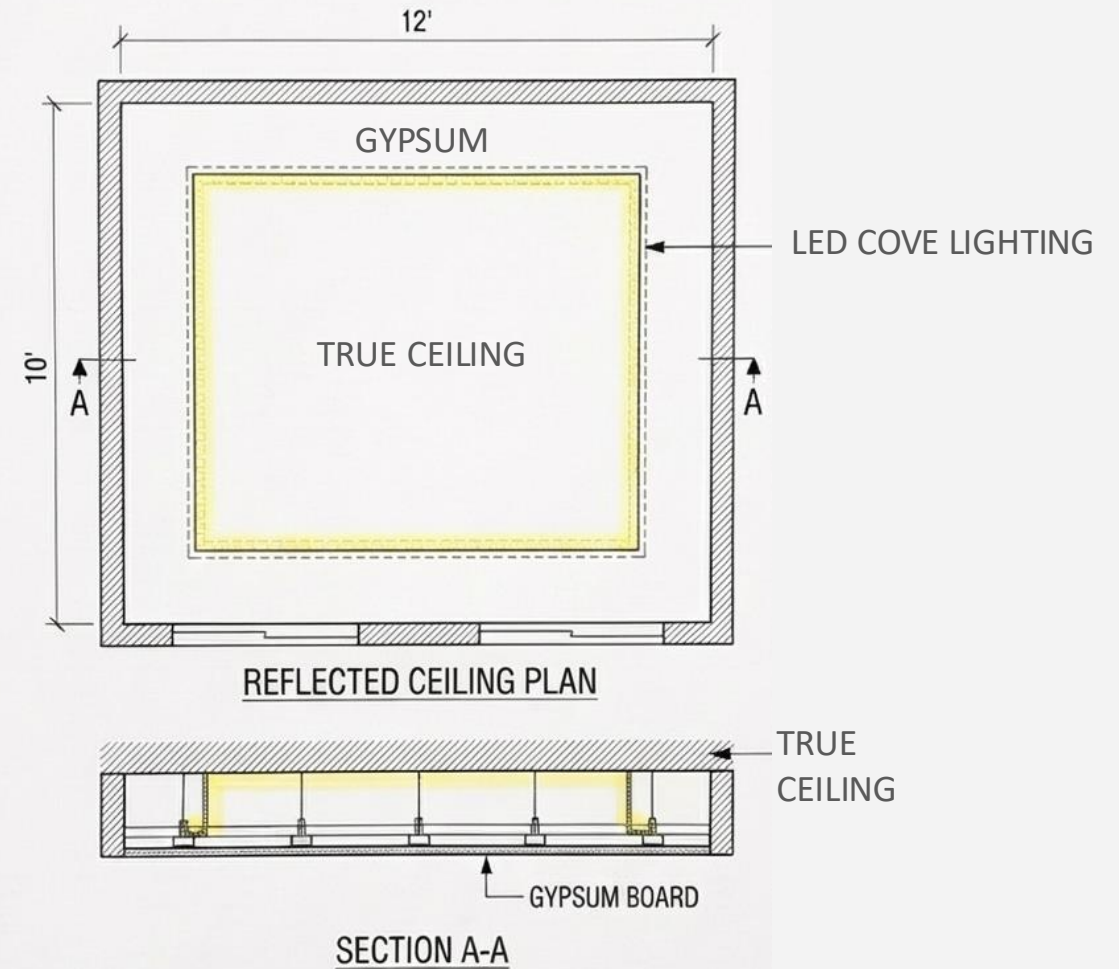
LED cost: $36 \times 120 = \mathbf{₹4,320}$ (From Electrical SKU)

Total approximate cost: ₹12,360

Note: Typical drop height: **4–6 inches**, depending on design and services

Drop height:

- Does not change area
- Does not affect costing in sq.ft
- Must be coordinated with AC ducts, beams, and slab level



1c. Plain Gypsum Ceiling (Without Cove)

- **Calculation of gypsum false ceiling (Rectangular type)**

Step 1: Calculate total ceiling area

Ceiling area is calculated based on **room size**, not design shape.

Formula:

Length × Width

Example:

Room size = **12 ft × 10 ft**

Total ceiling area =

$12 \times 10 = \mathbf{120 \text{ sq.ft}}$

Step 2: Rectangular drop understanding

Assume:

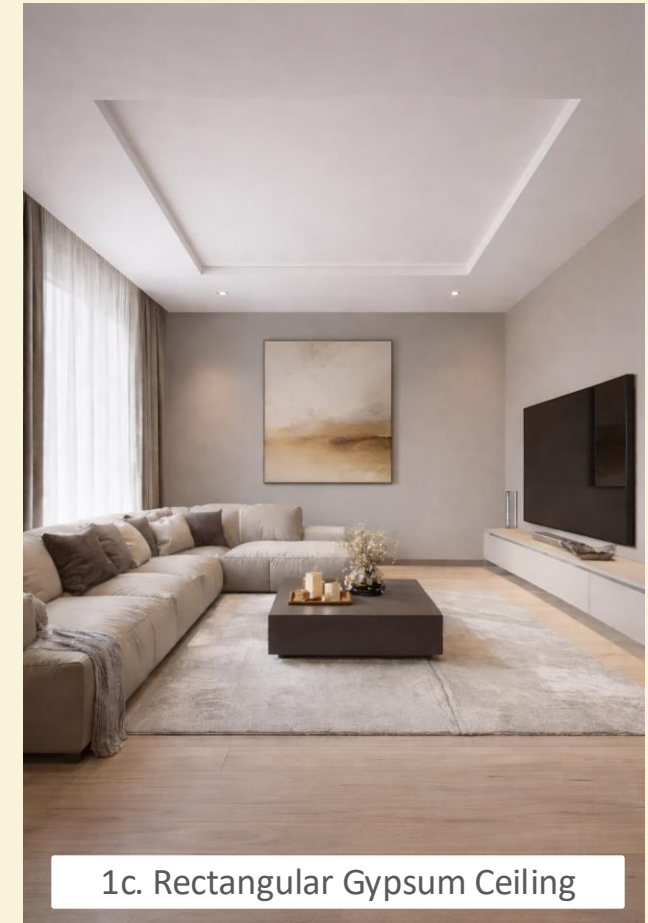
- Drop offset = **1 ft from all sides**
- Inner dropped rectangle size = **10 ft × 8 ft**

Inner dropped area =

$10 \times 8 = \mathbf{80 \text{ sq.ft}}$

Outer flat portion =

$120 - 80 = \mathbf{40 \text{ sq.ft}}$



1c. Rectangular Gypsum Ceiling

Important Note – Plain Gypsum False Ceiling

In Plain False Ceiling, **maximum 2 vertical steps** are allowed.

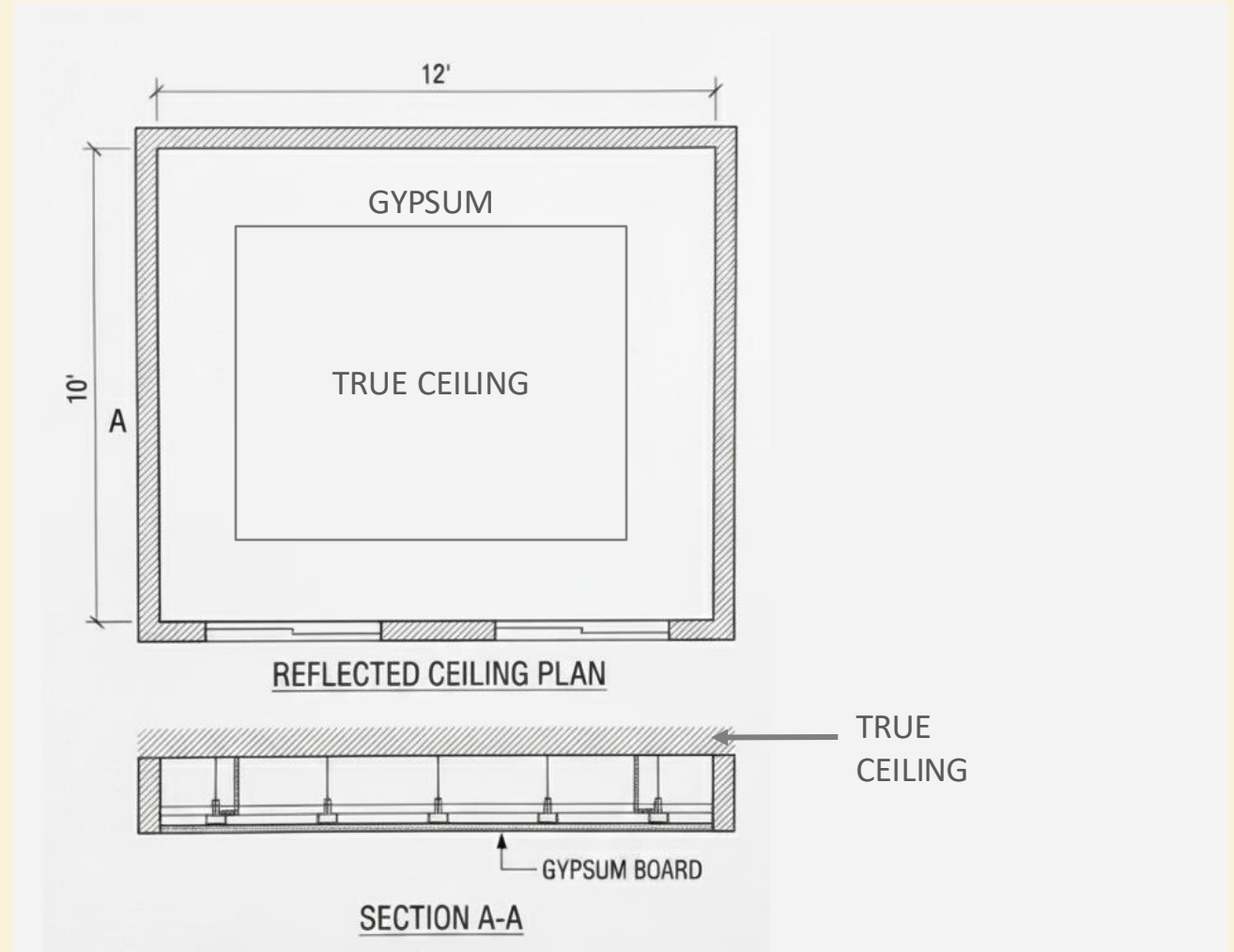
Step 3: Cost calculation (Gypsum only)

(Example rate – adjust as per company standard)

- Gypsum false ceiling (material + framework + finishing):
₹120 / sq.ft

Calculation:

$$40 \times 120 = \text{₹}4,800$$



Note: Typical drop height: **4–6 inches**, depending on design and services

Drop height:

- Does not change area**
- Does not affect costing in sq.ft**
- Must be coordinated with AC ducts, beams, and slab level

3a. Curved Gypsum Ceiling (With Cove)

- **Calculation of curved gypsum false ceiling** (Calculated as Rectangle)

Curved gypsum false ceilings are **not calculated by curve geometry**.

For execution and costing, they are **converted into an equivalent rectangle** using the **maximum effective length and width**.

Given dimensions

- Overall ceiling size: **21 ft × 11 ft**
- Ceiling runs along full length: **21 ft**

Width selection

The curve moves inside at different points:

- First curve: 1 ft inside
- Second curve: 3 ft inside
- Third curve: 1 ft 6 in inside

For calculation, **only the maximum usable width is considered**.

Effective width = 11 ft – 1 ft = **10 ft**



3a. Curved Gypsum Ceiling

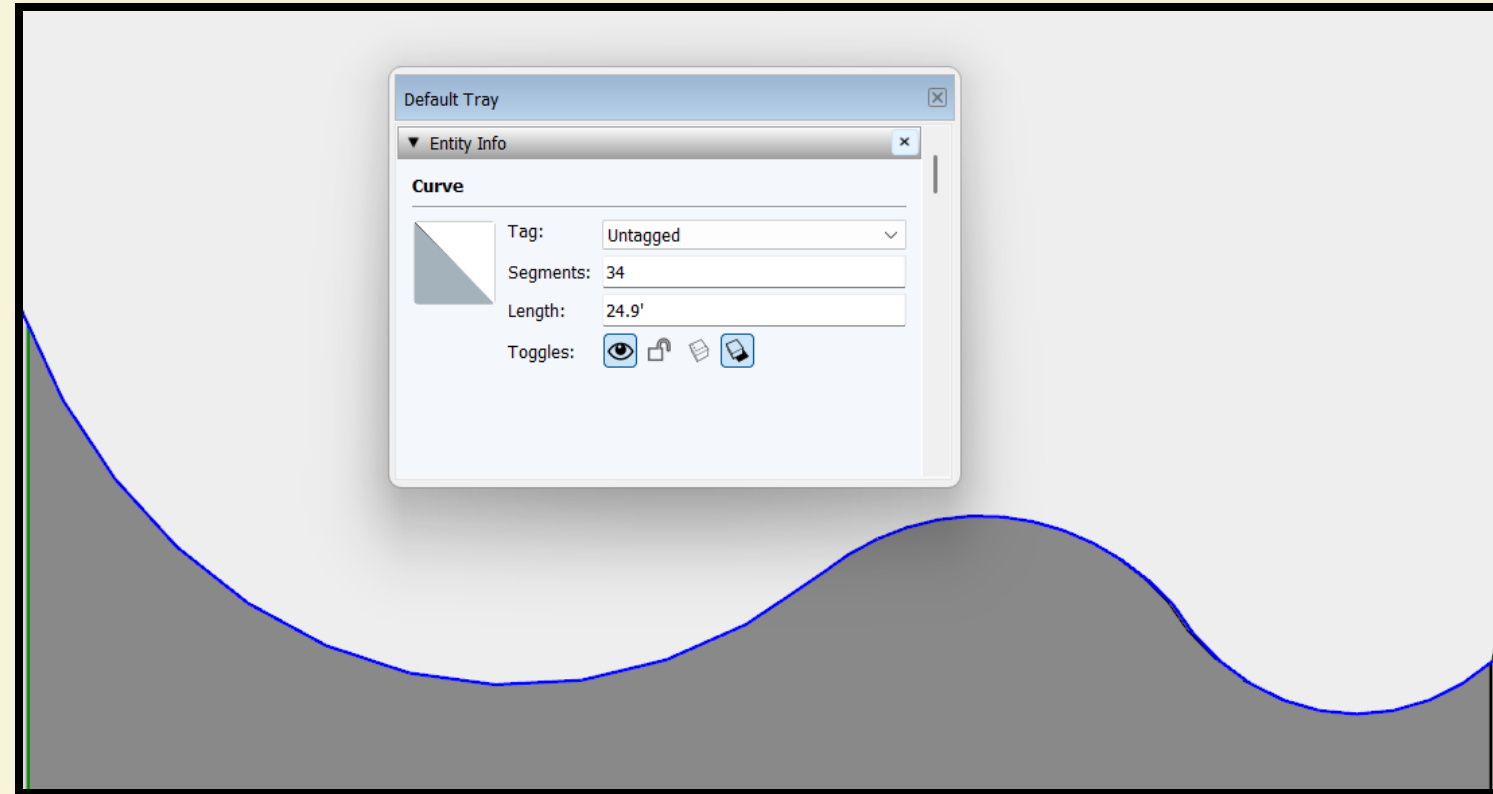
Important Note – Curved Gypsum False Ceiling
Only **1 vertical step** is allowed; multiple steps are not applicable

How to calculate curved (cove) length in SketchUp

- Select the curved line of the cove.
- Right-click → Make Group.
- Select the grouped curve.
- Right-click → Entity Info.
- Check the “Length” value.

The **Length** shown = curved cove length (rft)

Use this value directly for cove / curved false ceiling calculation.



Step 1: Calculate ceiling area

Formula:

Length × Width

Calculation:

$$21 \times 10 = \mathbf{210 \text{ sq.ft}}$$

5% Wastage to be considered for the designer false ceiling

$$210 \text{ sq.ft} + 5\% = \mathbf{220.5 \text{ sq.ft}}$$

Step 2: Cove lighting calculation

Cove lighting is provided **only along the curved front side.**

- Cove length = **24.9 rft**

Step 3: Cost calculation (example)

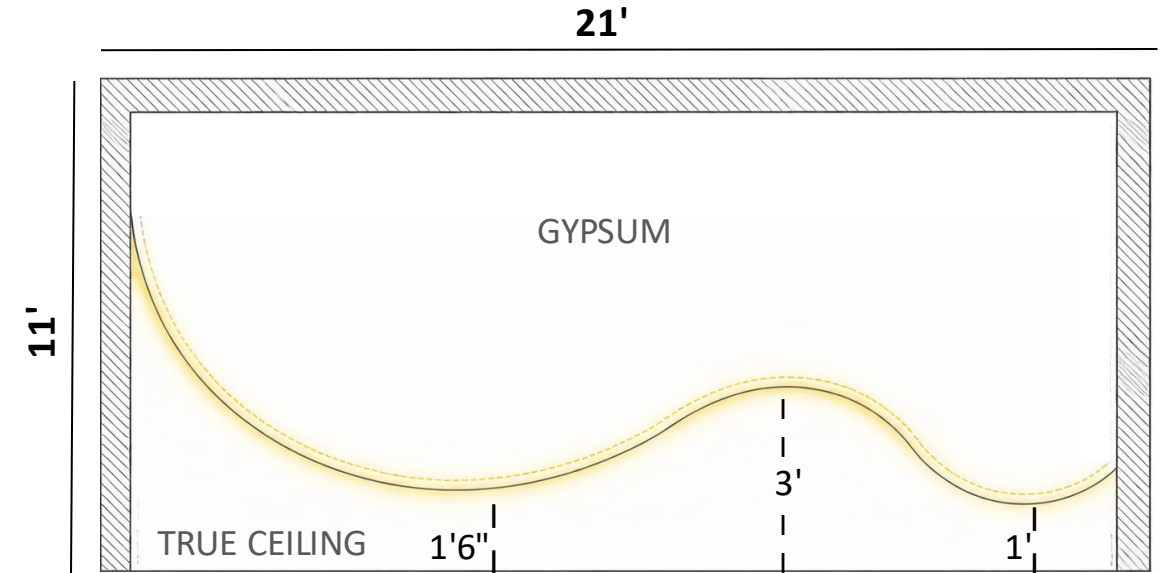
- Gypsum false ceiling: ₹120 / sq.ft
- Cove provision: ₹350 / running ft

Calculation:

$$\text{Gypsum ceiling cost: } 220.5 \times 120 = \mathbf{₹26,460}$$

$$\text{Cove cost: } 21 \times 350 = \mathbf{₹7,350}$$

$$\text{Total ceiling cost (approx.)} = \mathbf{₹33,810}$$



3a. Curved Gypsum Ceiling
with cove

3b. Curved Gypsum Ceiling (Without Cove)

- **Calculation of curved gypsum false ceiling** (Calculated as Rectangle)

Curved gypsum false ceilings are **not calculated by curve geometry**.

For execution and costing, they are **converted into an equivalent rectangle** using the **maximum effective length and width**.

Given dimensions

- Overall ceiling size: **21 ft × 11 ft**
- Ceiling runs along full length: **21 ft**

Width selection

The curve moves inside at different points:

- First curve: 1 ft inside
- Second curve: 3 ft inside
- Third curve: 1 ft 6 in inside

For calculation, **only the maximum usable width is considered**.

Effective width = 11 ft – 1 ft = **10 ft**



3b. Curved Gypsum Ceiling w/t Cove

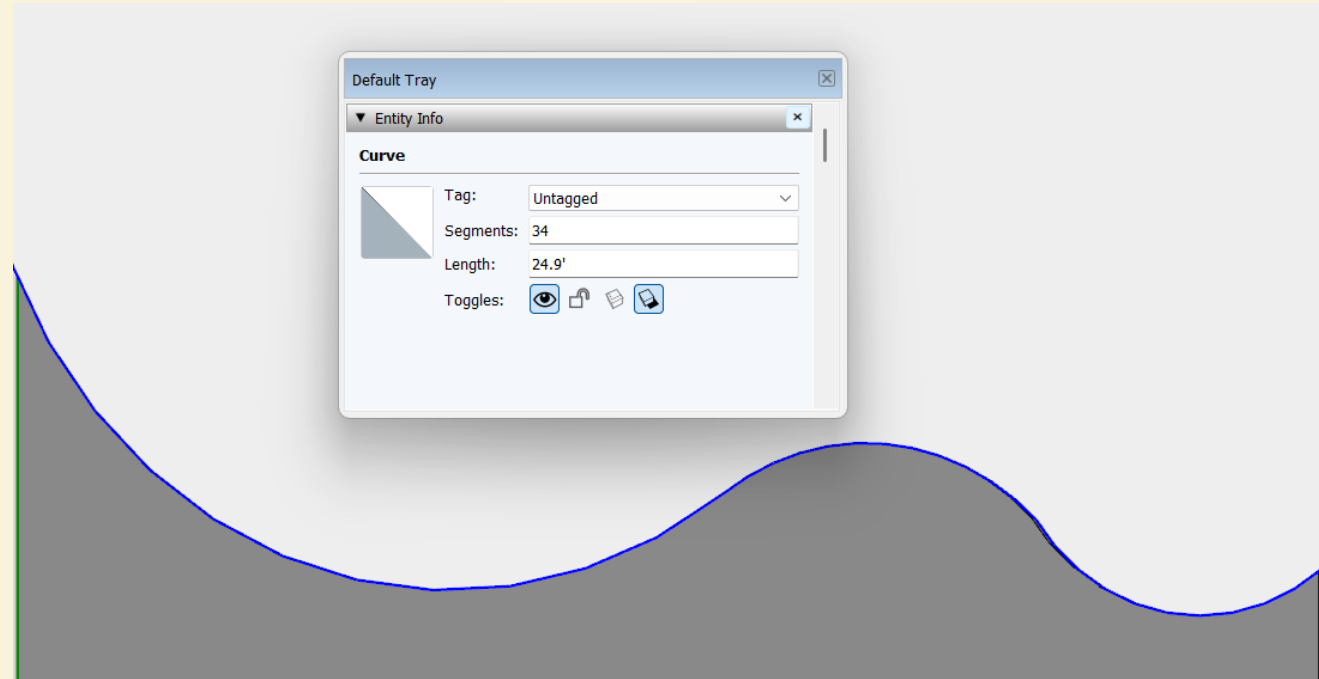
Important Note – Curved Gypsum False Ceiling
Only **1 vertical step** is allowed; multiple steps are not applicable

How to calculate curved (cove) length in SketchUp

- Select the curved line of the cove.
- Right-click → Make Group.
- Select the grouped curve.
- Right-click → Entity Info.
- Check the “Length” value.

The **Length** shown = curved cove length (rft)

Use this value directly for cove / curved false ceiling calculation.



Step 1: Calculate ceiling area

Formula:

Length × Width

Calculation:

$$21 \times 10 = \mathbf{210 \text{ sq.ft}}$$

5% Wastage to be considered for the designer false ceiling

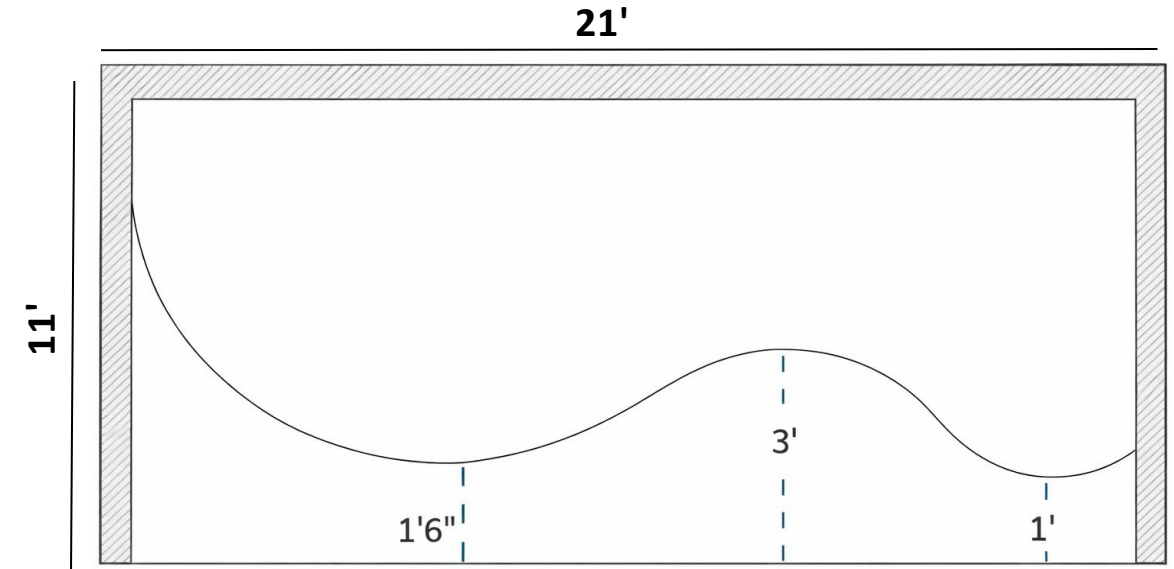
$$210 \text{ sq.ft} + 5\% = \mathbf{220.5 \text{ sq.ft}}$$

Step 2: Cost calculation (example)

- Gypsum false ceiling: ₹120 / sq.ft

Calculation:

$$\text{Gypsum ceiling cost: } 210 \times 120 = \mathbf{₹26,460}$$



3b. Curved Gypsum Ceiling
without cove

One Complete Example (Very Important)

Rectangular False Ceiling WITH Cove

- Horizontal area = 210 sq.ft
- Cove only on curved front (length = 21')

Entry in system:

- Horizontal Panel (sq.ft): **210**
- Vertical Panel – With Cove (rft): **21**
- Vertical Panel – Without Cove (rft): **0**

Rectangular False Ceiling WITHOUT Cove

- Length = 21'
- Width = 10'

Entry in system:

- Horizontal Panel (sq.ft): **210**
- Vertical Panel – Without Cove (rft): **21**
- Vertical Panel – With Cove (rft): **0**

Design Gypsum False Ceiling

Deal Price

- per nos

Total Price (with tax)

-

ERP Code

DesignGypsumFalseCeiling

Category

False ceiling

Horizontal Panel

Horizontal Panel (sft)

0

(Rs. 104.17)

Vertical Panel

Vertical Panel - With Cove upto 6 inch (rft)

0

(Rs. 266.0)

Vertical Panel - Without Cove (rft)

0

(Rs. 89.0)

Unit Price

0.0

Quantity

Add to Cart

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Gypsum False Ceiling – FAQs

1. How is the area of a gypsum false ceiling calculated?

The gypsum false ceiling area is calculated based on the **room's plan dimensions (Length × Width)**, irrespective of the ceiling design, steps, or drop levels.

2. Does the ceiling drop height affect the gypsum area or cost?

No. The typical drop height of **4–6 inches** does **not change the ceiling area** and **does not impact costing**, which is always calculated in square feet.

3. How many vertical steps are allowed in a plain gypsum false ceiling?

A **maximum of two vertical steps** is allowed in a plain gypsum false ceiling design.

4. How many vertical steps are allowed in a curved gypsum ceiling?

Only **one vertical step** is allowed in curved gypsum false ceiling designs. Multiple steps are not applicable.

5. How is a curved gypsum ceiling area calculated?

Curved ceilings are **not calculated using curve geometry**.

They are converted into an **equivalent rectangle** using the **maximum effective length and width** for execution and costing.

6. How is cove lighting length calculated?

Cove lighting is calculated in **running feet (RFT)** based on the **actual length where the cove is provided**, not the room perimeter.

7. How do we calculate curved cove length accurately?

Curved cove length is calculated using **SketchUp** by selecting the curved line, grouping it, and checking the **“Length” value** under Entity Info. This value is used directly in RFT.

8. Is the entire ceiling area charged when a partial drop or border is provided?

No. Only the **actual gypsum-covered area** (such as the border or dropped portion) is considered for gypsum costing, while the full room area is used for ceiling reference.

9. Why is wastage added for designer or curved ceilings?

Designer and curved ceilings involve additional cutting, shaping, and alignment. Hence, a **standard 5% wastage** is added to the calculated area.

10. Is wastage added for plain gypsum false ceilings?

Wastage is generally **not required** for simple plain ceilings unless specified by design or execution constraints.

11. Are LED strips included in gypsum ceiling costing?

No. LED strips are **part of the electrical scope** and are charged separately based on running feet.

12. Does providing a cove change the gypsum ceiling area?

No. Cove provision does **not alter the ceiling area**. Cove costs are calculated **separately in running feet**.

13. How should gypsum false ceiling quantities be entered in the system?

- **Horizontal Panel** → Area in sq.ft
- **Vertical Panel – With Cove** → Running feet
- **Vertical Panel – Without Cove** → Running feet

14. Can ceiling design be finalized after AC or electrical execution?

No. Gypsum ceiling design must be **finalized and frozen after coordination** with AC ducts, beams, and electrical layouts to avoid rework.

15. Are all images and drawings shown in the document final execution references?

No. All images and drawings are **for reference purposes only**. Final execution depends on **approved designs, site conditions, and coordination drawings**.

Muchas Gracias

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